

6.3 Transition Element Complexes: Isomers, Reactions & Stability (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	6. Inorganic Chemistry (A Level Only)
Topic	6.3 Transition Element Complexes: Isomers, Reactions & Stability (A Level Only)
Difficulty	Hard

Time allowed: 50

Score: /38

Percentage: /100

Question 1a

Carboplatin and oxaliplatin are both platinum(II) diamines, similar to cisplatin, which are widely used in chemotherapy to treat a wide variety of cancers. However, their use is often plagued with severe toxicity and the development of resistance, which leads to disease progression. Recently, oxoplatin, iproplatin, ormaplatin and satraplatin are Pt complexes that have been used clinically or in clinical trials.

The central platinum ions in the four complexes shown in **Fig. 1.1** all have the same shape and oxidation number.

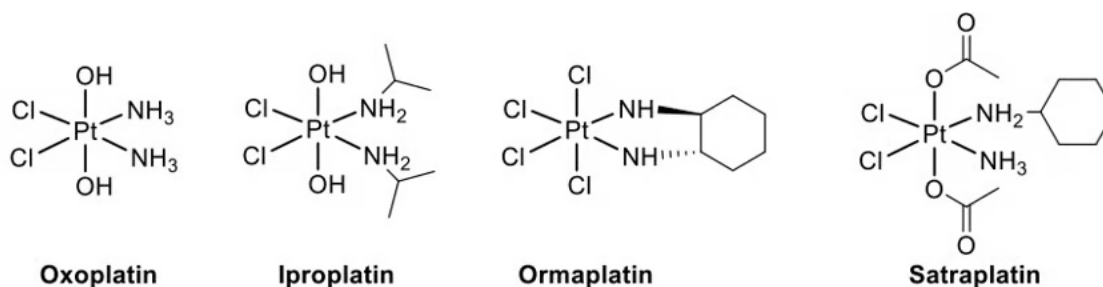


Fig. 1.1

Give the following information for the complexes.

oxidation number of the platinum ion

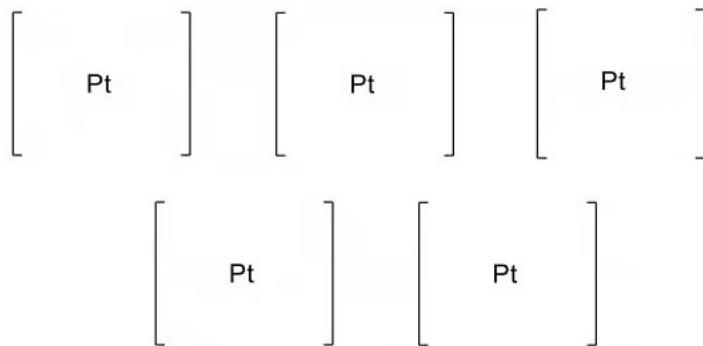
shape of the complexes

[2 marks]

Question 1b

One isomer of the $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2(\text{OH})_2]$ complex oxoplatin is shown in part (a) Fig. 1.1.

Complete three-dimensional diagrams to show the shape of the other oxoplatin stereoisomers and indicate the chiral one(s).



[4 marks]

Question 1c

Platinum complexes can be considered prodrugs that are intracellularly activated to kill cancer cells.

Cis-platin can be produced by the reduction of the prodrug *cis, trans, cis*- $[\text{PtCl}_2(\text{OCOCH}_3)_2(\text{NH}_3)_2]$, which has a similar structure to satraplatin.

Draw the structure of *cis, trans, cis*- $[\text{PtCl}_2(\text{OCOCH}_3)_2(\text{NH}_3)_2]$.

[1 mark]

Question 2a

When chromium(III) sulfate dissolves in water, a green solution containing the $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ ion forms.

i)

State the bond angles found in this complex ion.

[1]

ii)

Explain why the chromium(III) complex ion is coloured.

[3]

[4 marks]

Question 2b

Ethylenediaminetetraacetate, EDTA^{4-} , shown in Fig. 2.1. is a polydentate ligand.

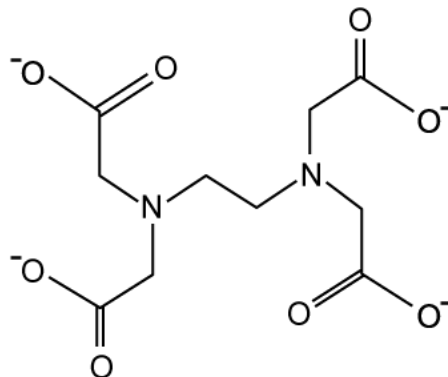
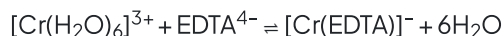


Fig. 2.1

When a solution of EDTA^{4-} is added to a solution of $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ ions, a new complex ion is formed.



i)

Name the type of reaction occurring here.

[1]

ii)

Write an expression for the stability constant, K_{stab} , of $[\text{CrEDTA}]^{-}$ in this reaction.

[1]

iii)

The numerical value of the K_{stab} of $[\text{CrEDTA}]^{-}$ is 2.51×10^{23} in this reaction.

Suggest what this indicates about the position and entropy of the equilibrium.

[3]

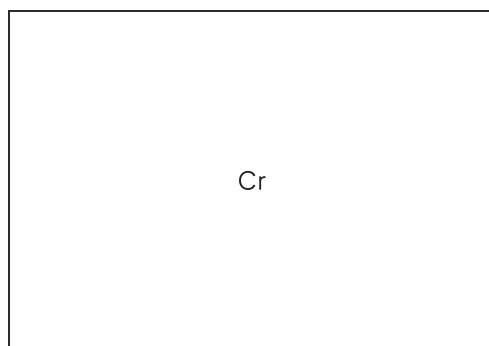
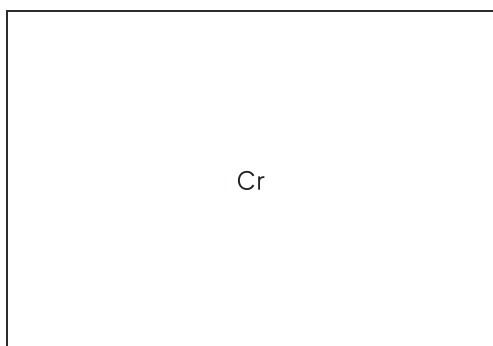
[5 marks]

Question 2c

Compound **L** is a complex with the empirical formula $\text{CrN}_4\text{H}_{12}\text{Cl}_3$

The formula of compound **L** contains one chloride ion and a complex ion **M**, which has two stereoisomers.

Complete three-dimensional diagrams to show the shape of the stereoisomers of complex ion **M**.



[3 marks]

Question 2d

Chromium(III) picolinate, shown in Fig. 2.2, is a neutral complex that can be prepared from the weak acid, picolinic acid.

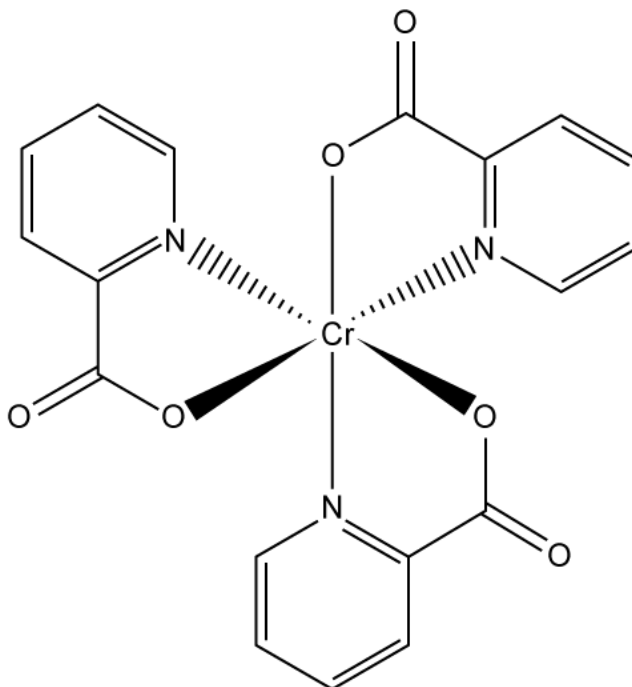


Fig. 2.2

Chromium(III) picolinate is used in tablets as a nutritional supplement for chromium.

- i)
Draw the structure of the ligand in chromium(III) picolinate.

[1]

- ii)
A typical tablet of chromium(III) picolinate contains 200 μg of chromium.

Calculate the mass, in g, of chromium(III) picolinate in a typical tablet. Give your answer to **three** significant figures.

[2]

[3 marks]

Question 3a

Iron(II) gluconate, $\text{C}_{12}\text{H}_{22}\text{FeO}_{14}$, is the active ingredient in some brands of iron supplements.

A student carries out an experiment to determine the mass of iron(II) gluconate in one tablet of an iron supplement, using the method below.

1. The student crushes two tablets and dissolves the powdered tablets in dilute sulfuric acid
2. The student makes up the solution from step 1 to 250.0 cm^3 in a volumetric flask.
3. The student then titrated 25.0 cm^3 portions of the solution obtained in step 2 with $0.00200\text{ mol dm}^{-3}$ potassium manganate(VII).

The student obtains a mean titre of 13.50 cm^3 .

In this titration, 1 mole of manganate(VII) ions reacts with 5 moles of iron(II) ions.

Determine the mass, in mg, of iron(II) gluconate in **one** tablet. Give your answer to 3 significant figures.

[4 marks]

Question 3b

The typical concentration of a potassium manganate(VII) solution used in redox titrations is $0.0200\text{ mol dm}^{-3}$.

Use the information in part (a) to explain, quantitatively, why the student replaced this with a $0.00200\text{ mol dm}^{-3}$ potassium manganate(VII) solution for this titration.

[2 marks]

Question 3c

Some iron supplements contain iron(II) sulfate or iron(II) fumarate.

Table 3.1 shows the information taken from the labels of two iron supplements, **A** and **B**.

Table 3.1

Iron supplement	Iron compound	Mass of iron compound in one tablet / mg
A	Iron(II) sulfate, FeSO_4	180
B	Iron(III) fumarate, $\text{C}_4\text{H}_2\text{FeO}_4$	210

State which iron supplement, **A** or **B**, would provide the greater mass of iron per tablet. Show your workings.

[1 mark]

Question 4a

Write an equation for the reaction between ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$, and sodium hydroxide, NaOH .

[1 mark]

Question 4b

15.00 cm^3 of $\text{H}_2\text{C}_2\text{O}_4(\text{aq})$ requires 10.30 cm^3 of a 0.25 mol dm^{-3} solution of NaOH for complete neutralisation using a phenolphthalein indicator for the first permanent colour change.

15.00 cm^3 of the same $\text{H}_2\text{C}_2\text{O}_4(\text{aq})$ required 12.35 cm^3 of potassium permanganate, KMnO_4 , solution for complete oxidation to carbon dioxide and water in the presence of dilute sulfuric acid to further acidify the $\text{H}_2\text{C}_2\text{O}_4(\text{aq})$ for the first permanent colour change.

Calculate the moles of $\text{H}_2\text{C}_2\text{O}_4(\text{aq})$ therefore the concentration of the solution.

[3 marks]

Question 4c

Write the full redox equation, including state symbols, for this redox titration.

[3 marks]

Question 4d

Calculate the concentration of the potassium permanganate, KMnO_4 , solution.

[2 marks]

Model Answers: Hard

1a

a) The correct information is:

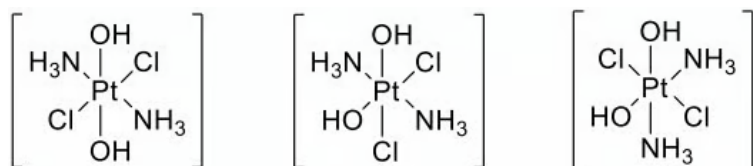
- Oxidation number of the platinum ion = +4; [1 mark]
- Shape of the complexes = Octahedral; [1 mark]

[Total: 2 marks]

- You are told in the question that all four complexes have the same oxidation number for the platinum ion and the same shape
- Oxoplatin is the simplest of the four complexes as it contains familiar ligands that you should know the charges for:
 - $\text{OH} = -1$
 - $\text{Cl} = -1$
 - $\text{NH}_3 = 0$
- The overall charge on the complex is zero meaning, which means that:
 - $\text{Pt} + (2 \times \text{OH}) + (2 \times \text{Cl}) + (2 \times \text{NH}_3) = 0$
 - $\text{Pt} + (2 \times -1) + (2 \times -1) + (2 \times 0) = 0$
 - $\text{Pt} + (-2) + (-2) = 0$
 - $\text{Pt} - 4 = 0$
 - $\text{Pt} = 4$
- All of the complexes have 6 coordinate bonds, which means that the shape must be octahedral

1b

b) The stereoisomers of oxoplatin are:



- Each correct structure; [1 mark]

The chiral isomers are:



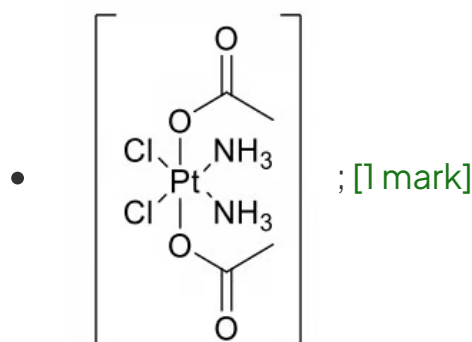
- Both enantiomer structures; [1 mark]

[Total: 4 marks]

- The stereoisomers can be identified by arranging the ligands in different positions relative to one another around the central Pt atom
- Careful:**
 - Do not give the oxoplatin structure from Fig. 1.1 as an answer, this will not score a mark
 - When drawing these isomers, it can be easy to draw the same structure twice but from a different angle
- You can see none of the top three stereoisomers are chiral by drawing a dotted line and flipping the structure
 - For each of these, you will be able to rotate this flipped structure to align with the original isomer drawn
 - Therefore, they are not non-superimposable images

1c

c) The structure of *cis, trans, cis*-[PtCl₂(OCOCH₃)₂(NH₃)₂] is:



[Total: 1 mark]

- There are three pairs of different ligands in this structure

- Cl
 - OCOCH_3
 - NH_3
- The *cis*, *trans*, *cis* in the complex formula *cis*, *trans*, *cis*- $[\text{PtCl}_2(\text{OCOCH}_3)_2(\text{NH}_3)_2]$ refers, in order, to the three pairs of ligands
 - *Cis* - Cl
 - *Trans* - OCOCH_3
 - *Cis* - NH_3
- **Remember**
 - *Cis* means that the ligands are adjacent / next to each other
 - *Trans* means that the ligands are opposite one another

2a

a)

i) The bond angles found in this complex are:

- 90° ; [1 mark]

ii) The chromium(III) complex is coloured because:

- The ligand / water molecule causes the d orbitals to split (into 2 energy levels); [1 mark]
- Light / energy (in the visible region) is absorbed to promote electrons (to the higher d orbitals); [1 mark]
- Remaining light / unabsorbed light / complementary colour / green light is transmitted; [1 mark]

[Total: 4 marks]

- The complex formed in $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ is octahedral as there are six identical, monodentate water ligands coordinately bonded to a chromium(III) ion
 - This shape results in a 90° bond angle between the water ligands
- The electrons in the five 3d orbitals have the same energy in the first-row transition metal atoms (Ti - Cu)
- When the atom forms a complex ion, the ligands cause the d orbitals to split into two groups, due to electrons in the ligand repelling electrons in 3d orbitals which are in close proximity
- You will lose a mark for saying that the remaining light is transmitted when the

electrons return to their ground state, as it is the complementary colour of light that you would observe in a complex ion

- Electrons returning to their ground state is used to explain the colour produced in flame test emission spectra not forming complex ions

2b

b)

i) The type of reaction is:

- Ligand exchange / displacement / replacement / substitution; [1 mark]

ii) The expression for the stability constant, K_{stab} , of $[\text{CrEDTA}]^-$ in the reaction is:

$$K_{\text{stab}} = \frac{[\text{CrEDTA}]^{2-}}{[\text{Cr}(\text{H}_2\text{O})_6]^{3+} [\text{EDTA}]^{4-}}; [1 \text{ mark}]$$

iii) The value of K_{stab} indicates that:

- The equilibrium lies far to the right-hand side / $[\text{Cr}(\text{EDTA})]^-$ is more stable (as the equilibrium constant is large)

OR

The forward reaction is very feasible / spontaneous / goes to completion; [1 mark]

- ΔS_{system} is positive / increases entropy; [1 mark]
- As 2 moles of reactants forms 7 moles of products

OR

There is an increase in the number of moles / particles / molecules; [1 mark]

[Total: 3 marks]

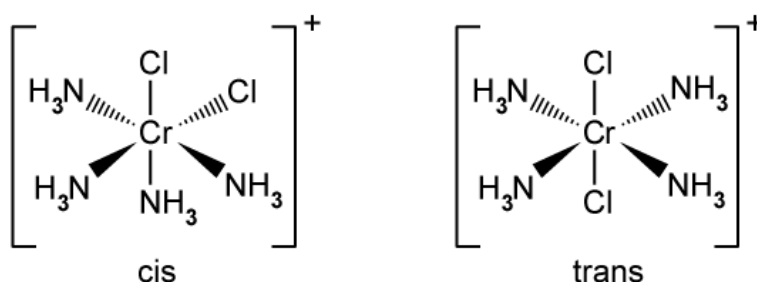
- The reaction equation given in the question shows that the EDTA^{4-} is replacing all of the water ligands
 - This is a clear ligand change reaction
- K_{stab} is essentially products (raised to appropriate powers) $\div$ reactants (raised to appropriate powers)
 - Remember:** The concentration of water is not included in the expression as the water is in excess
- The feasibility of any reaction can be explained by the magnitudes of the two

energy terms ΔH and ΔS

- When Cr^{3+} bonds with a molecule of EDTA, the enthalpy change for ΔH is small however the ΔS is very different:
 - The forward reaction has a high positive ΔS value (as the system has become more disordered) due to an increase in the number of moles of products compared to reactants
- **Careful:** Only referring to entropy to answer in your answer will not get full marks
- **Remember:** A high equilibrium constant indicates that the forward reaction is thermodynamically favoured and the equilibrium is favoured to the right hand (product side)

2c

c) The three-dimensional diagrams to show the shape of the stereoisomers of complex ion **M** are:



- 3D drawings each with 2 'out wedges', 2 'in wedges' and 2 lines in plane of paper
AND
Correct *cis* and *trans* labels; [1 mark]
- Structures show correct ligands (4 NH_3 **AND** 2 Cl); [1 mark]
- 1+ charge on **BOTH** structures; [1 mark]

[Total: 3 marks]

- If complex / compound **L** contains one of the chloride ions, then the complex ion **M** must have the empirical formula $\text{CrN}_4\text{H}_{12}\text{Cl}_2^+$
 - The N_4H_{12} is likely to be four ammonia ligands
 - This means that the likely structure of complex ion **M** is $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
- Complex ion **M** is an octahedral complex containing four ammonia ligands and two chloride ligands which means that it can exist as *cis* and *trans* isomers
 - The *cis*-isomer will have the two chloride ligands adjacent to each other,

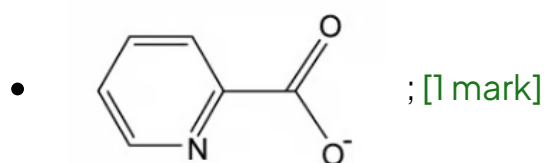
with a bond angle of 90° between the chromium–chloride coordinate bonds

- The *trans*-isomer will have the two chloride ligands opposite each other, with a bond angle of 180° between the chromium–chloride coordinate bonds
- **Careful:** The question asks for 3D drawings, so make sure you are confident drawing these complexes with the correct wedges in appropriate places

2d

d)

i) The structure of the ligand in chromium(III) picolinate is below:



ii) The mass, in g, of chromium(III) picolinate in a typical tablet is:

- Picolinate $M_r = 418.0$

AND

$$\text{Moles of chromium} = \frac{200 \times 10^{-6}}{52.0} = 3.85 \times 10^{-6}; \text{ [1 mark]}$$

- Mass = $3.85 \times 10^{-6} \times 418.0 = 1.61 \times 10^{-3} \text{ g}; \text{ [1 mark]}$

[Total: 3 marks]

- Don't be tempted to also draw a negative charge on the nitrogen in the picolinate ligand as well as on COO^-
 - Three out of the five nitrogen electrons are involved in bonding, leaving a lone pair of electrons which is used to form the coordinate bond with the chromium (III) ion
 - The nitrogen does not need a negative charge next to it as it is not a nitrogen ion, but a nitrogen atom with a lone pair
- **Remember:** $1 \mu\text{g} = 10^{-6} \text{ g}$
- **Careful:** Take your time working out the M_r of chromium(III) picolinate as this is where the most common error is made in this calculation

3a

a) The mass, in mg, of iron(II) gluconate in one tablet is:

- Moles of $\text{MnO}_4^- = \frac{13.50}{1000} \times 0.00200 = 2.7 \times 10^{-5}$; [1 mark]

- Moles of Fe^{2+} (in 25.0 cm^3) = $2.7 \times 10^{-5} \times 5 = 1.35 \times 10^{-4}$

AND

Moles of Fe^{2+} (in 250 cm^3) = $1.35 \times 10^{-4} \times 10 = 1.35 \times 10^{-3}$; [1 mark]

- M_r of $\text{C}_{12}\text{H}_{22}\text{FeO}_{14} = 445.8$

AND

Mass of $\text{C}_{12}\text{H}_{22}\text{FeO}_{14}$ in 2 tablets = $1.35 \times 10^{-3} \times 445.8 = 0.60183 \text{ g}$; [1 mark]

- Mass of $\text{C}_{12}\text{H}_{22}\text{FeO}_{14}$ in 1 tablet = 301 mg; [1 mark]

[Total: 4 marks]

- There are a few things to watch out for with this mass calculation question which may catch you out:
 - Don't misread the question and determine the mass of Fe instead of the mass of $\text{C}_{12}\text{H}_{22}\text{FeO}_{14}$ in the tablet
 - This would give you a final answer of 37.7 mg, not 301 mg
 - Take care calculating the M_r of $\text{C}_{12}\text{H}_{22}\text{FeO}_{14}$
 - $(12 \times 12.0) + (22 \times 1.0) + 55.8 + (14 \times 16.0) = 445.8$
 - Make sure you determine the mass of $\text{C}_{12}\text{H}_{22}\text{FeO}_{14}$ in one tablet and not two
 - The student originally crushes up two tablets as mentioned in step 1
 - $0.60183 \text{ g} \div 2 = 0.300915 \text{ g}$
 - Convert your mass from g into mg
 - $0.300915 \text{ g} \times 1000 = 300.915 \text{ mg}$
 - Quote your final answer to three significant figures
 - Do not state the calculator value of 300.915 mg because you may not get the final mark

3b

b) The student changed the concentration of the potassium manganate(VII) solution from this with a $0.0200 \text{ mol dm}^{-3}$ to this with a $0.00200 \text{ mol dm}^{-3}$ because:

- A $0.0200 \text{ mol dm}^{-3}$ potassium manganate(VII) solution would produce a titre of 1.35 cm^3 ; [1 mark]

- Which would be hard to read from a burette

OR

Which has a high (percentage) error / uncertainty; [1 mark]

[Total: 2 marks]

- **Careful:** This question tells you to use the information in part (a)
 - It does not tell you to use your answer to part (a)
 - This is a common mistake from students seeing "Use..... part (a)" and the "explain, quantitatively" in the question
- The information in part (a) tells you that a $0.00200 \text{ mol dm}^{-3}$ potassium manganate(VII) solution produces a 13.5 cm^3 titre
 - Since a $0.0200 \text{ mol dm}^{-3}$ potassium manganate(VII) solution is ten times stronger / more concentrated, it will require a tenth of the titre, i.e. $13.5 \div 10 = 1.35 \text{ cm}^3$
- This volume is unrealistic to measure with a standard 50.0 cm^3 burette
 - It would have a high percentage error / uncertainty

3c

d) The iron supplement which would provide the greater mass of iron per tablet is:

- **B**
AND
A: Mass of Fe = 66 mg
AND

3b

b) The student changed the concentration of the potassium manganate(VII) solution from this with a $0.0200 \text{ mol dm}^{-3}$ to this with a $0.00200 \text{ mol dm}^{-3}$ because:

- A $0.0200 \text{ mol dm}^{-3}$ potassium manganate(VII) solution would produce a titre of 1.35 cm^3 ; [1 mark]
- Which would be hard to read from a burette

OR

Which has a high (percentage) error / uncertainty; [1 mark]

[Total: 2 marks]

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- This volume is unrealistic to measure with a standard 50.0 cm^3 burette
 - It would have a high percentage error / uncertainty

3c

d) The iron supplement which would provide the greater mass of iron per tablet is:

- **B**
AND
A: Mass of Fe = 66 mg
AND
B: Mass of Fe = 69 mg; [1 mark]

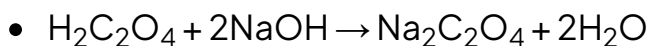
[Total: 1 mark]

- You have to substantiate your statement with the correct calculations here to get one mark

1. Calculate the M_r of each iron compound
 - **A** / $\text{FeSO}_4 = 55.8 + 32.1 + (4 \times 16.0) = 151.9$
 - **B** / $\text{C}_4\text{H}_2\text{FeO}_4 = (4 \times 12.0) + (2 \times 1.0) + 55.8 + (4 \times 16.0) = 169.8$
 2. Calculate the percentage of Fe in each compound
 - **A** / $\text{FeSO}_4 = 55.8 \div 151.9 \times 100 = 36.745\%$
 - **B** / $\text{C}_4\text{H}_2\text{FeO}_4 = 55.8 \div 169.8 \times 100 = 32.862\%$
 3. Apply these percentages to the mass of each compound in one tablet:
 - **A** / $\text{FeSO}_4 = 36.745 \div 100 \times 180 = 66.1 \text{ mg}$
 - **B** / $\text{C}_4\text{H}_2\text{FeO}_4 = 32.862 \div 100 \times 210 = 69.0 \text{ mg}$
- Therefore, table **B** contains more iron per tablet

4a

a) The equation for the reaction between ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$, and sodium hydroxide, NaOH is:



OR



[Total: 1 mark]

- The formula for ethanedioic acid is written in a bit of a confusing way in the question and you may be more familiar with $(\text{COOH})_2$
- This is a straightforward acid-base reaction where a salt and water are produced

4b

b) To calculate the moles of $\text{H}_2\text{C}_2\text{O}_4$ in the solution and therefore the concentration of the solution:

- Moles of $\text{NaOH} = \frac{0.25 \times 10.30}{1000} = 2.575 \times 10^{-3} \text{ moles}$; [1 mark]
- Moles of $\text{H}_2\text{C}_2\text{O}_4 = \frac{2.575 \times 10^{-3}}{2} = 1.2875 \times 10^{-3} \text{ moles}$; [1 mark]
- Concentration of $\text{H}_2\text{C}_2\text{O}_4 = \frac{1.2875 \times 10^{-3}}{15} \times 1000 = 0.0858 \text{ mol dm}^{-3}$; [1 mark]

[Total: 3 marks]

- Using your equation from part (a), you can deduce the moles and, therefore, the concentration of the ethanedioic acid solution
 - $\text{H}_2\text{C}_2\text{O}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{C}_2\text{O}_4 + 2\text{H}_2\text{O}$

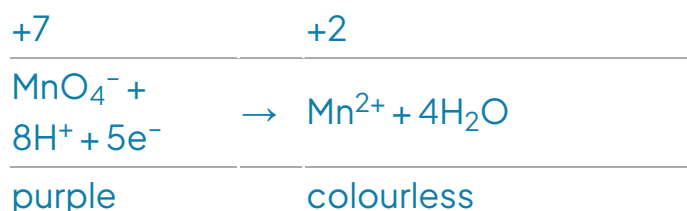
4c

c) The full redox equation is

- Half equation 1: $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{l})$; [1 mark]
- Half equation 2: $\text{H}_2\text{C}_2\text{O}_4 (\text{aq}) \rightarrow 2\text{CO}_2 (\text{g}) + 2\text{H}^+ (\text{aq}) + 2\text{e}^-$; [1 mark]
- Overall equation: $2\text{MnO}_4^- (\text{aq}) + 6\text{H}^+ (\text{aq}) + 5\text{H}_2\text{C}_2\text{O}_4 (\text{aq}) \rightarrow 2\text{Mn}^{2+} (\text{aq}) + 8\text{H}_2\text{O} (\text{l}) + 10\text{CO}_2 (\text{g})$; [1 mark]

[Total: 3 marks]

- If state symbols are not included or are incorrect, then a maximum of one mark will be lost
- In order to write the full equation from the two half equations, you will need to work out what the products of each half equation are as these are not given to you in the question
- The colour change when MnO_4^- ions are used in a redox titration is from purple to very pale pink / colourless



- In the question, it explains that CO_2 gas is released, therefore the half equation must be:
 - $\text{H}_2\text{C}_2\text{O}_4 \rightarrow 2\text{CO}_2 + 2\text{H}^+ + 2\text{e}^-$
- The final step is to form the full equation
 - Double the manganate equation:
 $2\text{MnO}_4^- + 16\text{H}^+ + 10\text{e}^- \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O}$
 - Multiply the ethanedioic acid equation by 5:
 $5\text{H}_2\text{C}_2\text{O}_4 \rightarrow 10\text{CO}_2 + 10\text{H}^+ + 10\text{e}^-$
 - Combine the half-equations:
 $2\text{MnO}_4^- + 16\text{H}^+ + 10\text{e}^- + 5\text{H}_2\text{C}_2\text{O}_4 \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} (\text{l}) + 10\text{CO}_2 + 10\text{H}^+ + 10\text{e}^-$

- Cancel out appropriate terms (hydrogen ions and electrons:



4d

d) To calculate the concentration of the potassium permanganate solution:

- Moles of KMnO_4 = moles of $\text{H}_2\text{C}_2\text{O}_4 \times \frac{2}{5} = 1.2875 \times 10^{-3} \times \frac{2}{5} = 5.15 \times 10^{-4}$; [1 mark]
- Concentration of $\text{KMnO}_4 = \frac{5.15 \times 10^{-4} \times 1000}{12.35} = 0.0417 \text{ mol dm}^{-3}$; [1 mark]

[Total: 2 marks]

- Using the full equation will help you calculate the concentration of KMnO_4
- You know the number of moles of $\text{H}_2\text{C}_2\text{O}_4$ and therefore will need to apply the ratio:
 - $2\text{MnO}_4^- + 6\text{H}^+ + 5\text{H}_2\text{C}_2\text{O}_4 \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 10\text{CO}_2$
 - The ratio of $\text{H}_2\text{C}_2\text{O}_4$ to MnO_4^- is 5:2
 - Therefore, you must multiply the moles of $\text{H}_2\text{C}_2\text{O}_4$ by $\frac{2}{5}$ to arrive at the correct value (or multiply by 2 and then divide by 5)

